

LIMITS AND CONTINUITY

$$\begin{aligned} \lim_{\theta \rightarrow 0} \frac{\cos(\theta) - 1}{\theta} &= \lim_{\theta \rightarrow 0} \frac{\cos(\theta) - 1}{\theta} \cdot \frac{\cos(\theta) + 1}{\cos(\theta) + 1} \\ \lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta} &= \lim_{\theta \rightarrow 0} \frac{\cos^2(\theta) - 1}{\theta(\cos \theta + 1)} \quad \begin{array}{l} \sin^2 \theta + \cos^2 \theta = 1 \\ \therefore \cos^2 \theta - 1 = -\sin^2 \theta \end{array} \\ &= \lim_{\theta \rightarrow 0} \frac{-\sin^2 \theta}{\theta(\cos \theta + 1)} \\ &= -\lim_{\theta \rightarrow 0} \sin \theta \cdot \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} \cdot \lim_{\theta \rightarrow 0} \frac{1}{\theta(\cos \theta + 1)} \\ &= -0 \cdot 1 \cdot \lim_{\theta \rightarrow 0} \frac{1}{\theta(\cos \theta + 1)} \\ &= 0 \end{aligned}$$

if f is differentiable at a ,
 f is continuous at a .

$$\begin{array}{ll} \text{if } f \text{ is differentiable at } a, & \text{if } f \text{ is continuous at } a, \\ \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = f'(a) & \lim_{x \rightarrow a} f(x) = f(a) \end{array}$$

take: $\begin{aligned} &\lim_{x \rightarrow a} f(x) - f(a) \\ &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} (x - a) \\ &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \cdot \lim_{x \rightarrow a} (x - a) \\ &= f'(a) \cdot 0 = 0 \end{aligned}$	Now, if: $\begin{aligned} &\lim_{x \rightarrow a} f(x) - f(a) = 0, \\ &\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} [f(x) - f(a) + f(a)] \\ &= \lim_{x \rightarrow a} [f(x) - f(a)] + \lim_{x \rightarrow a} f(a) \\ &= 0 + f(a) \\ &= f(a) \end{aligned}$ $\therefore f$ is continuous at a.
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$$e = \lim_{x \rightarrow 0} (1 + x)^{1/x} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$$

REMEMBER THESE

TRIG DERIVATIVES!

$$\frac{d}{dx} \sin x = \cos x$$

Let $f(x) = \sin x$

$$\begin{aligned} \therefore f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} \quad \sin(a+b) = \sin a \cos b + \sin b \cos a \\ &= \lim_{h \rightarrow 0} \frac{\sin x \cos h + \sin h \cos x - \sin x}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sin x (\cos h - 1) + \sin h \cos x}{h} \\ &= \lim_{h \rightarrow 0} \sin x \cdot \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} + \lim_{h \rightarrow 0} \frac{\sin h}{h} \cdot \lim_{h \rightarrow 0} \cos x \\ &= \sin x \cdot 0 + 1 \cdot \cos x \\ &= 0 + \cos x = \cos x = \text{RHS} \end{aligned}$$

$$\frac{d}{dx} \cos x = -\sin x$$

Let $f(x) = \cos x$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h} \quad \cos(a+b) = \cos a \cos b - \sin a \sin b \\ &= \lim_{h \rightarrow 0} \frac{\cos x \cos h - \sin x \sin h - \cos x}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cos x (\cos h - 1) - \sin x \sin h}{h} \\ &= \lim_{h \rightarrow 0} \cos x \cdot \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} - \lim_{h \rightarrow 0} \sin x \cdot \lim_{h \rightarrow 0} \frac{\sin h}{h} \\ &= \cos x \cdot 0 - \sin x \cdot 1 \\ &= 0 - \sin x = -\sin x = \text{RHS} \end{aligned}$$

$$\frac{d}{dx} \sec(x) = \frac{d}{dx} \left(\frac{1}{\cos x} \right)$$

$$\frac{d}{dx} \sec(x) = \sec(x) \tan(x)$$

$$\begin{aligned} &= \frac{d}{dx} (\cos x)^{-1} \\ &= -(\cos x)^{-2} \cdot \frac{d}{dx} (\cos x) \\ &= -\frac{1}{\cos^2(x)} \cdot (-\sin x) \\ &= \frac{1}{\cos(x)} \cdot \frac{\sin x}{\cos x} \\ &= \sec(x) \cdot \tan(x) \end{aligned}$$

$$\frac{d}{dx} \cot x = \frac{d}{dx} \left(\frac{1}{\tan x} \right)$$

$$\boxed{\frac{d}{dx} \cot x = -\csc^2 x}$$

$$= \frac{d}{dx} \left(\frac{1}{\frac{\sin x}{\cos x}} \right)$$

$$= \frac{d}{dx} \left(\frac{\cos x}{\sin x} \right)$$

$$= \frac{\frac{d}{dx}(\cos x) \cdot \sin x - \cos x \frac{d}{dx}(\sin x)}{\sin^2 x}$$

$$= \frac{-\sin x \cdot \sin x - \cos x \cdot \cos x}{\sin^2 x}$$

$$= \frac{-\sin^2 x - \cos^2 x}{\sin^2 x}$$

$$= \frac{-(\sin^2 x + \cos^2 x)}{\sin^2 x}$$

$$\sin^2 x + \cos^2 x = 1$$

$$= -\frac{1}{\sin^2 x} = -\csc^2 x$$

INVERSE TRIG DERIVATIVES:

63. $f(x) = \sin^{-1}(5x)$

65. $y = \tan^{-1}\sqrt{x-1}$

63. let $y = \sin^{-1}(5x)$

$$\Rightarrow \sin y = 5x, \quad y \in [-\pi/2, \pi/2]$$

using implicit differentiation:

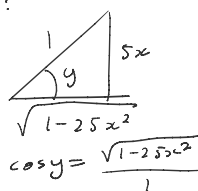
$$\frac{d}{dx} \sin y = \frac{d}{dx} 5x$$

$$\therefore \cos y \frac{dy}{dx} = 5$$

$$\therefore \frac{dy}{dx} = \frac{5}{\cos y}$$

$$= \frac{5}{\sqrt{1-25x^2}}$$

$$= \frac{5}{\sqrt{1-25x^2}}$$



$$\cos y = \frac{\sqrt{1-25x^2}}{1}$$

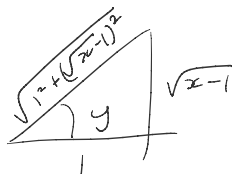
65. $y = \tan^{-1}(\sqrt{x-1})$

$\therefore \tan y = \sqrt{x-1} \quad y \in [-\pi/2, \pi/2]$

using implicit differentiation:

$\frac{d}{dx} \tan y = \frac{d}{dx} (\sqrt{x-1})$

$\therefore \sec^2 y \frac{dy}{dx} = \frac{1}{2\sqrt{x-1}}$



$\therefore \frac{dy}{dx} = \frac{1}{(2\sqrt{x-1}) \sec^2 y} \quad \cos y = \frac{1}{\sqrt{1+x-1}}$

$\therefore \frac{dy}{dx} = \frac{1}{(2\sqrt{x-1})(\sqrt{x})^2} = \frac{1}{\sqrt{x}}$

$\therefore \frac{dy}{dx} = \frac{1}{2x\sqrt{x-1}} \quad \therefore \sec y = \sqrt{x}$

69. $h(x) = (\arcsin x) \ln x$

let $y = \arcsin(x) \ln x$

using implicit differentiation:

$\frac{d}{dx} y = \frac{d}{dx} [\arcsin(x) \ln x]$

$\therefore \frac{dy}{dx} = \frac{d}{dx} [\arcsin(x)] \ln x + \arcsin(x) \frac{d}{dx} [\ln x]$

$= \frac{1}{\sqrt{1-x^2}} \cdot \ln x + \frac{\arcsin x}{x}$

$= \frac{\ln x}{\sqrt{1-x^2}} + \frac{\arcsin x}{x}$

QUESTION 23 [2 marks]

Show how you use the definition of the arctan = $\tan^{-1}$ function to determine the value of $\sin(2 \tan^{-1}(x))$.

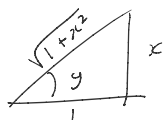
let $y = \tan^{-1}(x)$

$\Rightarrow \tan y = x, \quad x \in [-\pi/2, \pi/2] \quad \tan y = x = \frac{x}{1}$

$\therefore \sin 2y = 2 \sin y \cos y$

$\therefore \sin 2y = 2 \cdot \frac{x}{\sqrt{1+x^2}} \cdot \frac{1}{\sqrt{1+x^2}}$

$= \frac{2x}{1+x^2}$



$\sin y = \frac{x}{\sqrt{1+x^2}}$

$\cos y = \frac{1}{\sqrt{1+x^2}}$

$\therefore \sin(2 \tan^{-1}(x)) = \frac{2x}{1+x^2}$

LOG DERIVATIVES:

$$f(x) = \log_b x$$

$$\text{let } y = \log_b x$$

$$\therefore b^y = x$$

Using implicit:

$$\frac{d}{dx}(b^y) = \frac{d}{dx}(x)$$

$$\therefore b^y \ln(b) \cdot \frac{dy}{dx} = 1$$

$$\therefore \frac{dy}{dx} = \frac{1}{b^y \ln(b)}$$

$$b^y = x$$

$$\therefore \frac{dy}{dx} = \frac{1}{x \ln(b)}$$

if we sub in $b=e$:

$$f(x) = \log_e x$$

$$\therefore \frac{dy}{dx} = \frac{1}{x \ln(e)} \\ = \frac{1}{x}$$

$$\frac{d}{dx} \log_b x = \frac{1}{x \ln(b)}$$

$$\frac{d}{dx} \ln x = \frac{1}{x}$$

PRODUCT RULE:

$$\frac{d}{dx} [f(x) \cdot g(x)] = f'(x)g(x) + f(x)g'(x)$$

$$\frac{d}{dx} [f(x) \cdot g(x)] = \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) + [f(x)g(x+h) - f(x)g(x+h)] - f(x)g(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{g(x+h)[f(x+h) - f(x)] + f(x)[g(x+h) - g(x)]}{h}$$

$$= \lim_{h \rightarrow 0} g(x+h) \cdot \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} + \lim_{h \rightarrow 0} f(x) \cdot \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h}$$

$$= g(x) \cdot f'(x) + f(x) \cdot g'(x)$$

$$= f'(x)g(x) + f(x)g'(x) = \text{RHS}$$